ACCESS · INDIGENOUS SKILLS 4 SUCCESS

2027 program research workbook

Eleven programs. Nine skills. Practical tasks learners can try, check and explain.

Initial research draft · 22 October 2026 · Cohort dates tentative

Choose a starting point

Program briefs, worked examples and a pilot lesson, shaped by the ACCESS Numeracy Hub.

Start here
Nine-skills framework
Numeracy Hub direction
Welding
Asphalt Paving
Ironworker
Carpentry
Piping
Tower Crane
Marine Mechanical Technician
Security
Metal Fabrication
Educational Assistant
Nursing
11 numeracy starters
Paper-glider pilot lesson
Source and schedule audit
Research sources and next steps

ACCESS | Indigenous Skills 4 Success

2027 program research and learning design

Working draft · 22 October 2026 · All cohort dates remain tentative

Build from a real task. Show the steps. Practise together. Check the result. Explain where the skills were used.

This first research pack connects all 11 proposed cohorts to the nine current Skills for Success: **Reading, Writing, Numeracy, Digital, Problem solving, Communication, Collaboration, Adaptability, and Creativity and innovation**. It follows the practical workbook approach of ACCESS Trades Numeracy, reviewed at commit `7c0ceaab485e36aa45c596ac58e43bba63e4f824`.

Start here

Browse the styled workbook · Download the printable PDF

Resource	Use it for
Program briefs	One integrated task, nine observable skill applications, entry support and research questions for every cohort.
Worked numeracy starters	Eleven fictional practice examples with units, answers, checks and extension prompts.
Nine-skills teaching framework	Consistent skill names, evidence and a repeatable lesson pattern.
Numeracy Hub review	What the existing Hub actually does, reusable topic references and the direction for future materials.
Paper-glider pilot lesson	A proposed 60-minute adaptation of the photographed ACCESS lesson with all nine skills made visible.
Source and schedule audit	Date conflicts, missing information, the historical skills crosswalk and verification priorities.
Research sources and verification queue	What was inspected, what remains unverified, and the next primary sources to consult.

What is established so far

- **Reviewed directly:** the supplied program reference, the three visible lesson pages, and the Numeracy Hub repository, including its brand guidance and instructional structure.
- **Proposed here:** learning challenges, skill evidence, scaffolds, numeracy examples and the pilot lesson. These are curriculum-development drafts rather than provider-approved assessments.
- **Awaiting external verification:** current program prerequisites, credits, code/manual editions, certification dates, purchasing lists and clinical/licensing rules. Requests to the official research sites received a network-proxy `403 Forbidden`; the environment network changes were saved as a draft but have not been confirmed active.

- **Awaiting partner confirmation:** actual ACCESS course codes, delivery hours, dates, provider-specific requirements and funded supports. Public reference programs do not establish the content of a dedicated ACCESS cohort.

The tentative program sequence

Program	Supplied dates in 2027	Supplied partner / note	Brief
Welding	Jan 25–Mar 19	UAPIC	Open
Asphalt Paving	Feb 1–Mar 12	Six-week program; provider unspecified	Open
Ironworker	Mar 15–May 7	BCIT	Open
Carpentry	Mar 22–May 14	BCIT	Open
Piping	May 17–Jul 9	UAPIC	Open
Tower Crane	Jul 5–Aug 27	“IOUE Maple Ridge”; identity to confirm	Open
Marine Mechanical Technician	Jul 19–Sep 10	BCIT	Open
Security	Sep 14–Nov 6	TBD	Open
Metal Fabrication	Sep 14–Nov 6	BCIT	Open
Educational Assistant	Nov 8–Dec 17	TBD	Open
Nursing	TBD	BCIT; delivery model to confirm	Open

The source's summer paragraph gives July 19 for Tower Crane, conflicting with July 5 in its table and program section. Keep July 5 as the **table's provisional value**, with the conflict visible until resolved. Security and Metal Fabrication begin on a Tuesday and end on a Saturday; confirm that this pattern is intended.

Build the next teaching materials in this order

1. **Confirm scope and sources.** Establish the actual course/level and approved assessment outcomes for each cohort. Resolve the Tower Crane date and pathway questions first because the supplied reference flags a possible regulatory change.
2. **Pilot common foundations.** Use the paper-glider lab to observe learners' strategies, then select numeracy support from the existing Hub. Treat prior experience as a strength; offer private, practical feedback.
3. **Develop Welding and Asphalt first.** These are the earliest tentative starts. Obtain one approved work document per program and expand each starter into a complete learner sheet, worked example, instructor notes and short assessment.
4. **Extend the same pattern to the remaining cohorts.** Map each finished activity to a specific current competency after the provider confirms the course scope. Use skills tags consistently, without treating a tag as evidence of proficiency.
5. **Review with ACCESS and instructors.** Check relevance, accessibility, community preferences, technical accuracy, permissions for materials and assessment expectations before learner delivery.

The reference Numeracy Hub checkout was used for inspection. This research pack is maintained in the Business repository; publishing a live site is a separate step.

Nine skills, visible in the work

Back to the research pack

Proposed teaching framework · 22 October 2026

The current nine names below are also explicitly recorded in the Numeracy Hub's brand guidance. The descriptions here are working teaching prompts, not quotations from or a validated assessment instrument issued by ESDC. Verification against the current official framework is queued in the source register.

Skill	Ask the learner to…	Look for this evidence
Reading	Find and use an instruction, requirement or relevant detail.	The detail is identified and correctly applied; the learner can point to its source.
Writing	Make a record another person can use.	A clear, accurate note, form or explanation with the necessary units, dates or context.
Numeracy	Estimate, select a method, calculate or measure, and check.	Working, labelled units, a result and a reasonableness check.
Digital	Use a digital tool to find, organize, create or share task information.	An accurately created/retrieved file, table or record; the learner explains a relevant tool choice.
Problem solving	Identify a discrepancy, consider options and justify a response.	A reasoned decision supported by task evidence rather than a guessed answer.
Communication	Explain, listen, clarify and check shared understanding.	A useful exchange and a check-back that reveals whether the message was understood.
Collaboration	Agree roles, coordinate contributions and give useful feedback.	Shared decisions and an identifiable contribution from each learner.
Adaptability	Respond to a changed condition and revise the approach.	A revised plan and an explanation of what changed and why.
Creativity and innovation	Generate, try and improve a useful idea within the constraints.	A proposed improvement, a small test or comparison, and reflection on the result.

Document literacy remains central: learners read drawings, interpret tables, complete forms and navigate records. Integrate these tasks across Reading, Writing, Numeracy and Digital. Continuous learning runs through practice, reflection and transfer; it is not simply a new name for Adaptability. See the full historical crosswalk.

A repeatable lesson pattern

Use the photographed lesson's instructor/learner/resource/time structure and the Hub's visible worked steps. A program can demonstrate all nine across a sequence; select two or three focus skills for detailed feedback in an individual short lesson.

Stage	Instructor action	Learner action	Evidence
Connect	Introduce a believable task and ask about relevant experience.	Describe a familiar strategy or ask a question; personal disclosure is optional.	Starting strategy and support preferences.
Read the task	Supply a short document or clearly labelled diagram.	Identify givens, constraints, units and the required output.	Annotated task or explained interpretation.
Estimate and model	Demonstrate one example, with method and tool steps visible.	Predict an answer or range, then follow and question the method.	Estimate and method choice.
Practise	Offer a new example and purposeful roles.	Solve, measure, record and check; explain to a partner.	Task record, practical result and check-back.
Change one condition	Introduce a stated variation.	Revise the plan, compare options and test an improvement.	Revised work and justification.
Reflect and transfer	Ask what helped and where the approach applies next.	Name the skill used and point to evidence; choose a next step.	A short reflection tied to actual work.

Learner-sheet template: task → what you need → givens/diagram → estimate → method → worked example → your turn → check → changed condition → explain → next task.

Instructor-sheet template: purpose → provider outcome/source (pending until verified) → focus skills → materials → timed actions → anticipated misconceptions → worked answers → observation prompts → supports used → follow-up.

Feedback and access

Use simple task-specific observations: **not observed yet**, **completed with support**, **completed independently**, and **explained or transferred to a new case**. These are local feedback descriptors, not official Skills for Success levels, admission decisions or trade certification.

Record what support was used. Demonstration, a glossary, enlarged diagrams, assistive technology, extra practice and calculator access can make a task accessible. If Writing is the assessment target, an oral response can show understanding but does not by itself demonstrate Writing. Likewise, paper-based work can preserve the activity when a device is unavailable, but Digital remains unobserved until the learner uses a digital tool.

Ask learners how they prefer to participate. Use relevant community and workplace contexts with ACCESS/community input; do not invent shared Indigenous traditions or require disclosure of family, cultural or traumatic experiences. Offer collaborative improvement goals and private feedback. Use fictional records for practice involving students, clients, patients or incidents.

What the Numeracy Hub contributes

- Keep dimensions, units, drawings and calculator steps close to the question.
- Provide a worked example and a chance to try again with changed values.
- Show why an answer is reasonable, including common unit or decimal errors.
- Offer print-friendly work as well as digital practice.

- Start with approachable tasks and increase the reasoning demand deliberately. Use “Start”, “Practise” and “Extend” for this draft; formal complexity labels need a validated mapping.
- For later visual materials, carry forward warm cream, white task surfaces, red headings/actions and gold rules/highlights. Use the existing IS4S assets appropriately; do not add invented cultural motifs.

Technical and professional criteria come from the actual provider's current approved materials. Classroom arithmetic, communication practice and scenario work do not independently establish safe performance on equipment, permission to practise or qualification credit.

Numeracy Hub review for ACCESS IS4S program research

Reviewed 22 October 2026. Source review of `GOSTROGIS/access-trades-numeracy` at commit `7c0ceaab485e36aa45c596ac58e43bba63e4f824`. Checkout: `/workspace/reference-repos/access-trades-numeracy`. The reference checkout was left unchanged. This review inspected source and content; it does not claim a browser usability or runtime test.

What the hub actually establishes

The README describes math practice for ACCESS pre-apprentices in fabrication, pipe trades and millwright work. It includes a browser hub, Shape Canvas, calculator emulators, a textbook, exercise book, answer key, BC Building Code drawings, and carpentry slides. The README documents offline use and browser-local progress.

Its explicit layout direction is a **shop workbook**: diagram on layout paper beside numbered steps, then practice cards, with sources near the end (hub source, line 5). This is the strongest direction to carry into the new research pack: a practical task, an example, practice, feedback, and a clear workplace purpose.

The brand notes, line 29 name all nine current Skills for Success: Reading, Writing, Numeracy, Digital, Problem Solving, Communication, Collaboration, Adaptability, Creativity and Innovation. They give a numeracy description: “find, understand, use and report mathematical information presented through words, numbers, symbols and graphics.” They do **not** provide a researched description or assessment framework for the other eight skills. Their appearance in the brand notes is not evidence that the hub already assesses all nine.

Visual and writing direction observed

Element	Observed direction	Application to program research
Ground and cards	Warm ground `#F7F4F1`, white cards, ink `#1A1512`, slate `#4A4E54`, restrained borders, 6px corners	Plain document/workbook structure; readable program sections and task sheets
Colour	ACCESS red `#D2181F`, deep red `#C60016`, gold `#FFBC00`; 4px gold header rule	Red headings/actions, gold rules and highlight fills; never gold body text
Typography	Archivo for headings, IBM Plex Sans for prose, IBM Plex Mono/tabular figures for measurements	Retain clear headings and aligned units/data; the brand file calls this pairing recommended, not formally confirmed by ACCESS
Diagram treatment	White drawing paper remains white in dark mode; dimensions and geometric elements are labelled	Keep job sheets, diagrams and examples legible in print and on screen
Voice	Direct verbs, short explanations, actual materials and consequences: “On the job,” “Worked example,” “Practice tasks,” “Your answer”	Write “Read the drawing. Mark the length. Check the unit. Explain your choice.”
Feedback	“Right,” “Not yet,” and hints about the specific error	Feedback identifies an actionable next step, rather than labelling the learner

Evidence: brand colours, type guidance, hub tokens, diagnostic feedback.

The book wording checks also show a deliberate plain-language voice: no filler, no em dashes or semicolons in learner reading text, and no “as an AI” language. These are evidence about that project's style, not mandatory rules for a staff research register. Research uncertainty must remain explicit even though final learner materials exclude draft placeholders.

Teaching structure and interactions worth carrying forward

- Start with **On the job**: why this action matters in this occupation.
- Show a labelled diagram, form, table, schedule, or model, then number the steps.
- Include a worked example with answer and units. Calculator key sequences are visible and can run on the selected emulator. A global control hides operations so learners can attempt the task first (header and controls).
- Move into practice with new values. Guided practice gives the first steps and fades them across questions (exam settings).
- Give a reasoned correction: wrong unit, decimal position, sign, or premature rounding (feedback implementation).
- Revisit topics across days. The review exam uses missed topics and time since last practice; progress is local to that browser (topic record).
- Provide printable blank and answer versions: the hub offers Word and PDF exports (exports).
- Preserve accessibility choices: visible focus, reduced-motion handling, keyboard tabs, text-size controls and readable body text. Source presence is not a full accessibility audit.

The hub tags numeracy tasks Level 1–3 and separately offers Easy, Medium, Hard, Brutal and Nightmare settings (difficulty implementation). Do not present these local settings as validated proficiency levels for all nine skills. For the new program pack, a proposed progression of **Supported practice → Independent practice → Transfer to a new situation** is clearer and avoids importing the intimidating labels. Formal proficiency alignment requires a separate framework check.

Existing content paths for all 11 programs

The following are genuine existing numeracy starting points; the occupation-specific integrations are proposed. Tab fragments refer to the downloaded page `numeracy-hub/index.html`. The shell routing accepts `#shape`, `#ooo`, `#vlm`, `#angles`, `#fab`, `#pipe`, `#units`, `#mech`, `#coords`, `#measure`, and `#exam`. Module anchors inside embedded frames are internal identifiers, not additional outer-page deep links. Do not publish invented compound fragments such as `#fab/m-tape`.

Program	Existing resource and stable identifiers	Practical research/use direction
Welding	`numeracy-hub/index.html#fab`: internal `m-tape`, `m-square`, `m-nesting`; `#vlm`: internal `m-area` (area/perimeter)	Read a cut list, calculate lengths or weld perimeter, check diagonals, record fit-up measurements. Add an oral handover and a written discrepancy note.

Program	Existing resource and stable identifiers	Practical research/use direction
Asphalt Paving	`#vlm` : internal `m-vol` , `m-area` ; `#units` ; book chapter 11	Reuse area × thickness as a geometry exercise and convert units. Asphalt density, compaction, temperature and field specifications must come from the selected provider, not the hub's concrete example.
Ironworker	`#fab` : `m-weight` , `m-square` , `m-scale` ; `#mech` : levers and pulleys	Estimate mass from a supplied classroom density; read a plan; compare diagonals. Do not turn ideal mechanical-advantage examples into lift authorization or substitute them for approved load charts.
Carpentry	`#vlm` : internal `m-layout` and `m-area` ; `#measure` : rules/tapes, squares/levels; `carpentry/intake-142/PowerPoint/` , `carpentry/intake-142/PDF/`	Read scales, make cut lists, check square, estimate materials; adapt known carpentry slides only after checking their intake and code edition.
Piping	`#pipe` : `m1` Measuring and pipe lengths; `#units` ; pipe code-book/materials/glossary/drawings sections	Read a schematic, calculate cut lengths, find a table/Article, explain a material choice. The code section explicitly names NPC 2025; establish the applicable B.C. edition with the instructor before using it as a compliance reference.
Tower Crane	`#mech` : `m2` Levers, `m3` Pulleys; `#fab` : `m-weight` ; `#units`	Prepare the concepts of mass, units, radius and reading tables. Build communication/stop-and-clarify exercises around instructor-supplied simulation documents. No crane capacity can be established from these general numeracy modules.
Marine Mechanical Technician	`#mech` : `m4` Gears/sprockets/belts, `m6` Wheel and axle/torque, `m7` Hydraulics; `#measure` : calipers/micrometers/gauges	Read a mock service job, measure a part, compare a provider-supplied tolerance, calculate a ratio and document the finding. Manufacturer service limits remain the authoritative input.
Security	book chapter 1, Time on the job, tables/graphs from line 395	Use elapsed time, 24-hour clocks and a fictional patrol log. Pair observations with factual report writing, clarification and handover. The hub contains no BST/legal curriculum.
Metal Fabrication	`#fab` : `m-tape` , `m-stretch` , `m-nesting` , `m-weight` , `m-square` , `m-bending` , `m-cones` , `m-scale` ; `#coords` ; `#measure`	Closest existing match. Reuse layout, nesting and scrap tasks as a basis for design choices, team verification and a revision record. Shop-specific bend parameters must be supplied by the instructor.
Educational Assistant	chapter 1, fractions, Kitchen Math	Adapt scheduling, counting, proportions and reading charts to a fictional classroom activity; design more than one representation. Do not claim the trade hub supplies inclusion, behaviour-support or practicum standards.
Nursing	decimals, percent/ratio, units, chapter 1 time	Use general unit reasoning, estimates, elapsed time and data tables as preparation. Clinical medication tasks, notation and acceptable methods require nursing-faculty review; trade calculators/materials do not establish clinical competency.

The outer tab list is visible at hub lines 1078 onward. Embedded module payloads are at Volume/Layout L1241, Fabrication L1243, Pipe L1244, Mechanical L1246, and Measuring L1248.

Concrete examples that show the approach

1. **Ordering brackets:** 3 units × 14 brackets + 6 spares = 48. Boxes hold 10, so 48 ÷ 10 = 4.8 and the order is 5 boxes. The explanation distinguishes rounding purchases up from rounding possible cut pieces down. This creates a natural nine-skills extension: read the job, calculate, record the order, enter a spreadsheet, explain the rounding, peer-check, handle a quantity change and compare an alternative. Existing example.
2. **Timesheets:** 45 min ÷ 60 = 0.75 h. The lesson first explains clocks/elapsed time, then decimal hours and pay. It does not assume that a learner who reads decimals already understands that time is base 60. Useful for all 11 programs, particularly security, care and classroom schedules. Existing sequence.
3. **Proportion with a reasonableness check:** 18 m of pipe takes 4 h, so a same-rate classroom model gives 45 m in 10 h. The text says to keep units aligned and check that more work takes more time. Do not treat this simplified model as a production or staffing promise. Existing practice.
4. **Documents as tools:** the piping code-book section asks learners to locate a Division, read a reference number, find a table value, then do the math. This already combines reading, digital navigation and numeracy. It is an excellent pattern for security procedures, classroom plans or a simulated service manual, with authoritative content supplied for each occupation.

Research sources already named in the hub

These URLs were extracted from the repository. Their presence is verified; their current live contents and suitability for a 2027 cohort have **not** been independently verified in this review.

Source	Existing URL	Best use
Government of Canada, Skills for Success	https://www.canada.ca/en/services/jobs/training/initiatives/skills-success/understanding-individuals.html	Verify current nine-skill definitions and subcomponents
BCcampus, Math for Trades 2	https://opentextbc.ca/mathfortrades2/chapter/order-of-operations/	Open numeracy explanation and worked-example research
BCcampus, volume of cylinder	https://opentextbc.ca/mathfortrades2/chapter/volume-of-a-cylinder/	Volume and capacity lesson research
BCcampus, Pattern Development	https://opentextbc.ca/patterndevelopment/chapter/frustum-of-a-cone/	Fabrication drawings and pattern-development research
Camosun / BCcampus, Trades Access Common Core C-1	https://collection.bccampus.ca/textbook/BzEe9fcq	Hand tools and measuring; hub labels it CC BY 4.0, which still needs source-level confirmation before reuse
NIST SP 811, conversions	https://www.nist.gov/pml/special-publication-811/nist-guide-si-appendix-b-conversion-factors/nist-guide-si-appendix-b8	Authoritative unit factors; not occupation licensing or code rules

Source	Existing URL	Best use
B.C., Plumbing Code bulletin P24-01	https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/construction-industry/building-codes-and-standards/bulletins/p24-01_2024_bc_plumbing_code_revised_2024-05-28.pdf	Establish jurisdiction and historical 2024 adoption context, then check newer notices
Texas Instruments, TI-30Xa guide	https://education.ti.com/en/guidebook/details/en/E4CC4C51FCA94B33A48137F1FCE16670/30xa	Confirm calculator operations
Mitutoyo, measuring instruments	https://www.mitutoyo.com/webfoo/wp-content/uploads/E11003-Quick-Guide.pdf	Instrument-specific reading/care research

Some hub sources are commercial explainers, Wikipedia, old manuals or dealer-hosted copies. The reference hub is a useful lead set, not a substitute for provider curricula, current regulations or manufacturer instructions.

Proposed common program research structure

Use the same sequence for every program: **Workplace purpose → Cohort facts and open questions → Authentic document/task → Worked example → Supported practice → Independent attempt → Change in the situation → Reflection and transfer → Nine-skill evidence → Sources and instructor checks.**

Keep learner pages short and practical. Put admissions, credentials, purchasing questions and source-status notes in a staff research page. For each skill, name what the learner actually does and what the instructor can see or collect. A tag alone is not integration. Use two or three focus skills in a short lesson; make all nine observable over the module or sequence rather than forcing nine separate exercises into every activity.

Retain ACCESS/IS4S brand assets according to the brand-use notes. No repository-wide LICENSE or AGENTS.md was found in this checkout. Reusing the user's design direction does not establish redistribution rights for external code books, textbook passages, fonts or manufacturer images. Link or paraphrase with source attribution until the particular item's reuse terms are checked; original diagrams and activity wording can follow the existing style.

Welding — plan and document a practice assembly

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Welding”, and have not been independently confirmed.

Supplied context: January 25–March 19, 2027; Trades; UAPIC. Confirm the actual eight-week scope and assessment outcomes separately from a full foundation program. **Integrated challenge:** In pairs, prepare a materials plan and job record for an instructor-selected, non-structural practice assembly; explain the drawing, check quantities and revise after an approved design change. **Inputs to request:** Approved drawing, sample job card, material list, measurement tools and vocabulary sheet; process and safety instructions supplied by the welding instructor.

Skill	Proposed activity → observable evidence
Reading	Locate dimensions, drawing notes and relevant instructions → annotated drawing with the source of each required detail marked.
Writing	Complete a job card with sequence, observations and questions → clear, dated record distinguishing observations from assumptions.
Numeracy	Measure practice pieces and total cut lengths using one unit system → calculation sheet with units and a reasonableness check.
Digital	Enter quantities in a simple materials spreadsheet and save the correct drawing revision → named file with checked entries and revision identification.
Problem solving	Detect a mismatch between the drawing and materials list → explanation of the mismatch and the question that must be resolved before work.
Communication	Explain the planned sequence and request clarification of an unfamiliar symbol → partner can restate the task and unresolved issue.
Collaboration	Rotate reader, measurer and checker roles → jointly checked plan plus a note of each learner's contribution.
Adaptability	Respond to an instructor-issued dimension change → updated quantities and explicit identification of affected steps.
Creativity and innovation	Suggest a clearer job-card layout or material-identification method → small prototype and explanation of how it reduces confusion.

Accessible entry / stretch: Begin with a full-size diagram, labelled sample and whole-number measurements; extend to mixed measurements, an extra component or comparing two permitted stock layouts. **Evidence bundle:** Annotated drawing, quantity sheet, job card, revision note and learner reflection identifying one transferable skill.
Research/provider questions: Which process, materials, joint types and drawing conventions are in the cohort? Which texts/modules and PPE are supplied? What practical assessments and any progression credit are explicitly included?

Start from the Numeracy Hub

`numeracy-hub/index.html#fab` (tape measurements, checking square, nesting) and `#vlm` (area/perimeter). Start with repeated cut lengths, then build the job-card and drawing-reading tasks around them.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Confirm the actual eight-week welding processes and assessments. Retrieve the current Welder outline/profile and an approved drawing/symbol sample. Resolve which books and any testing are included; the historical foundation references are background.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Asphalt Paving — prepare a simulated small-site materials plan

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Asphalt Paving”, and have not been independently confirmed.

Supplied context: February 1–March 12, 2027; Trades; provider not identified. A public asphalt program is a reference, not confirmation of this six-week cohort's content. **Integrated challenge:** Use an instructor-approved site sketch to estimate a practice paving area, reconcile fictional delivery tickets and brief a team about a weather-related scheduling change. **Inputs to request:** Simplified site drawing, fictional delivery tickets, approved calculation assumptions, equipment vocabulary and a provider-authored weather scenario.

Skill	Proposed activity → observable evidence
Reading	Find dimensions and delivery details across a plan and tickets → highlighted fields linked to the materials plan.
Writing	Record delivery discrepancies and a revised work sequence → concise log that records facts and flags unanswered questions.
Numeracy	Calculate area and, where taught, volume using an instructor-supplied depth → worked estimate with consistent units; use supplied factors only for further conversions.
Digital	Enter ticket quantities in a table and compare planned with recorded totals → saved reconciliation sheet with checked formulas.
Problem solving	Investigate why two tickets do not match the planned order → identify plausible causes and information needed before choosing an explanation.
Communication	Give a short change-of-plan briefing using plain language → listener accurately identifies the changed time and their next task.
Collaboration	Allocate planner, ticket checker and briefing roles → shared plan with a completed peer check.
Adaptability	Reorder classroom planning tasks after a weather or delivery update → revised sequence that preserves instructor-specified constraints.
Creativity and innovation	Design a clearer ticket-checking aid → trial checklist that helps another learner find a planted discrepancy.

Accessible entry / stretch: Start with rectangular areas, colour-coded tickets and a calculator; extend to composite shapes or comparing quantities across several deliveries. **Evidence bundle:** Site calculation, delivery log, discrepancy explanation, revised schedule and brief-back observation. **Research/provider questions:** Who delivers the cohort, and what fills its six weeks? What equipment practice and transport are available? Which current specifications, reference factors and assessment criteria may be used in teaching?

Start from the Numeracy Hub

`numeracy-hub/index.html#vlm` (area and volume) and `#units`. Reuse the geometry and unit-checking approach. Actual asphalt properties and site specifications must come from the provider.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Identify the actual provider and the six-week breakdown. Check the current Asphalt Laydown Operator outline/profile and provider entry route. Obtain fictional tickets and permitted teaching assumptions before developing a materials-planning lesson.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Ironworker — reconcile a drawing and component inventory

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Ironworker”, and have not been independently confirmed.

Supplied context: March 15–May 7, 2027; Trades; BCIT. Confirm whether this cohort concerns Generalist, Reinforcing, introductory exposure or another defined scope. **Integrated challenge:** Assemble a tabletop model from an instructor-selected drawing, reconcile labelled model components and produce a handover listing unresolved discrepancies. **Inputs to request:** Approved simplified drawing, revision history, model components, fictional inventory and provider-selected terminology. Exclude live lifting or structural decisions from this classroom task.

Skill	Proposed activity → observable evidence
Reading	Match component marks and drawing notes to the inventory → marked drawing and correctly identified model parts.
Writing	Document missing or inconsistent components → discrepancy report with location, identifier and required clarification.
Numeracy	Count pieces and compare model dimensions with the drawing → checked quantity-and-measurement table with units.
Digital	Locate the assigned drawing revision and update a component register → correctly named file and revision recorded in the register.
Problem solving	Resolve a model-assembly mismatch by checking source documents → evidence trail identifying whether information is missing or inconsistent.
Communication	Practise an instructor-approved confirmation exchange for the tabletop task → clear request, repeat-back and confirmation observed.
Collaboration	Coordinate model assembly and independent checking → agreed roles and an identified correction from peer review.
Adaptability	Incorporate a revised component list without losing the original record → versioned list showing what changed and why.
Creativity and innovation	Propose a more readable model-part labelling system → test labels another learner can interpret accurately.

Accessible entry / stretch: Start with large printed identifiers and a small model; extend to a larger inventory or reconciling two drawing revisions. **Evidence bundle:** Component register, tabletop model, discrepancy report, revision record and communication observation. **Research/provider questions:** Which trade pathway and competency outcomes apply? Which drawing conventions and current manuals are used? What rigging, height-work and practical instruction are included under qualified supervision?

Start from the Numeracy Hub

`numeracy-hub/index.html#fab` (weight, checking square and scale). Use component inventories and model drawings to practise document interpretation and checks.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Confirm Generalist versus Reinforcing and the selected training level. Retrieve that exact outline/profile and current drawings/manuals. Verify the supplied reference to the 2026 outline change before citing it as current.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Carpentry — estimate and revise a small project

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Carpentry”, and have not been independently confirmed.

Supplied context: March 22–May 14, 2027; Trades; BCIT. Confirm the cohort's project and credit separately from foundation or apprentice technical training. **Integrated challenge:** Develop a cut list and materials estimate for an instructor-approved small practice project, build a paper/card model and explain the effect of a client-requested change. **Inputs to request:** Approved drawing, stock sizes, fictional prices, cut-list template and instructor-defined allowances; any actual tool work follows the partner's teaching plan.

Skill	Proposed activity → observable evidence
Reading	Extract sizes and notes from a project drawing and materials list → annotated plan with corresponding parts identified.
Writing	Write a short work sequence and clarification request → ordered steps and a question precise enough for the instructor to answer.
Numeracy	Calculate lengths, counts and a materials subtotal → checked cut list and estimate with units and stated allowances.
Digital	Use a spreadsheet to recalculate a fictional estimate → saved sheet that updates correctly when one quantity changes.
Problem solving	Identify why the available stock cannot satisfy the first cut plan → revised plan using approved stock and allowance rules.
Communication	Explain the project estimate and confirm the requested change → accurate summary of agreed dimensions and unresolved details.
Collaboration	Pair a planner with an independent checker, then swap → documented correction and shared final plan.
Adaptability	Revise the list after a change in size or available stock → traceable before-and-after quantity comparison.
Creativity and innovation	Compare two permitted layouts for clarity or material use → sketches and a reasoned choice within the design requirements.

Accessible entry / stretch: Offer a labelled model, measurement demonstration and a partly completed cut list; extend to fractions, a second stock length or comparing total costs. **Evidence bundle:** Cut list, calculation sheet, model, change note and explanation of a checking strategy. **Research/provider questions:** What project, tools and assessment tasks are planned? Which measurement conventions and code edition apply? Which books and digital access are actually required for this intake?

Start from the Numeracy Hub

`numeracy-hub/index.html#vlm` (layout and area), `#measure`, and the existing `carpentry/intake-142/` slides. Check the original intake and code edition before adapting slides.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Confirm the project, foundation/introduction/apprentice scope and outcomes. Retrieve the current Carpenter outline and provider-selected code edition. Verify the 2027 book package rather than reusing the supplied Fall 2026 purchase example.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Piping — interpret a route and prepare a materials list

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Piping”, and have not been independently confirmed.

Supplied context: May 17–July 9, 2027; Trades; UAPIC. Confirm the balance of Plumbing, Steamfitting and Sprinklerfitting and any stated credit for this cohort. **Integrated challenge:** Interpret a simplified instructor-approved pipe route, create a tabletop model and prepare a parts list; revise it after an approved route change. **Inputs to request:** Schematic, labelled sample fittings or cards, manufacturer/provider reference extracts and instructor-supplied measurement conventions. Use no live pressure, gas or connected systems.

Skill	Proposed activity → observable evidence
Reading	Match symbols, fitting labels and schematic notes → annotated route showing which source identifies each component.
Writing	Record a parts request and a route-change note → complete identifiers, quantities and clear description of the change.
Numeracy	Measure route segments and count fittings; apply instructor-taught allowances only where provided → checked length and quantity schedule.
Digital	Find an assigned part in an approved digital catalogue → saved reference with the correct identifier and relevant specification marked.
Problem solving	Identify a missing specification or incompatible model component → clear stop-and-clarify question supported by the references.
Communication	Describe the proposed route to a partner and confirm terminology → partner reconstructs the intended tabletop route accurately.
Collaboration	Share schematic reading, model assembly and checking → jointly reconciled model and parts list.
Adaptability	Update the model after the instructor changes one route segment → revised drawing and list agree with each other.
Creativity and innovation	Improve a parts-identification or route-explanation aid → prototype tested for clarity by another learner.

Accessible entry / stretch: Begin with colour-coded fittings and direct lengths; extend to an additional branch or instructor-selected offsets after the necessary teaching. **Evidence bundle:** Annotated route, parts list, measurement sheet, tabletop model and change record. **Research/provider questions:** Which trades, materials and joining methods are included? What codes/modules are supplied? What registration, supervision and assessment arrangements apply to any workplace component?

Start from the Numeracy Hub

`numeracy-hub/index.html#pipe` (Measuring and pipe lengths) and `#units`. The existing pipe material mentions NPC 2025; verify the applicable B.C. code and provider materials before using code-based tasks.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Confirm which of Plumbing, Steamfitting and Sprinklerfitting are included. Retrieve the applicable outlines separately. Check the exact credit, code/module list and registration arrangements rather than applying full-foundation claims to eight weeks.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Tower Crane — check a simulated planning information pack

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Tower Crane”, and have not been independently confirmed.

Supplied context: July 5–August 27, 2027; Trades; partner recorded as “IOUE Maple Ridge.” Verify the partner name, course level and the supplied document's July 2027 certification-transition claim. **Integrated challenge:** Review a provider-authored fictional planning pack, identify missing or conflicting information and deliver a clear hold-for-clarification handover. The task does not approve or execute a lift. **Inputs to request:** Trainer-approved scenario, explicitly educational chart extracts, equipment terminology, communication protocol and known-answer checking sheet.

Skill	Proposed activity → observable evidence
Reading	Locate the specified row, column, units and notes in an educational chart → accurately annotated extract, including conditions the learner must clarify.
Writing	Record missing information and who needs to clarify it → concise issue log without inventing operational values.
Numeracy	Practise unit conversion and checking supplied quantities in a classroom example → workings with units, independent check and clear identification of uncertainty.
Digital	Select the assigned scenario revision and complete its planning checklist → saved record tied to the correct source version.
Problem solving	Recognize a conflict between two fictional records → explanation of why the task must pause and which qualified person should resolve it.
Communication	Use provider-taught closed-loop communication in a role-play → clear message, accurate repeat-back and confirmation.
Collaboration	Rotate information checker, communicator and recorder roles → a shared issue list with individual contributions observed.
Adaptability	Respond to an instructor-issued change in conditions → withdraw the previous classroom plan and list what must be reassessed.
Creativity and innovation	Improve the readability of the training checklist → prototype that makes missing information easier to notice without altering technical rules.

Accessible entry / stretch: Begin with one chart row and a terminology key; extend to reconciling several documents and explaining why an apparently complete pack is insufficient. **Evidence bundle:** Annotated educational chart, calculation practice, issue log and observed role-play; no operational approval inferred. **Research/provider questions:** Is the partner IUOE Local 115, and which level is offered? How do the July 2027 rules affect intake and placement? Which registration, sponsor and qualified-supervision requirements apply?

Start from the Numeracy Hub

`numeracy-hub/index.html#units`, `#fab` (weight) and `#mech` (levers/pulleys) support basic concepts. They do not establish crane capacity or replace provider-controlled operational training.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Resolve July 5 versus July 19, the “IOUE” partner spelling, level structure and any sponsor/registration requirements. Independently verify the supplied July 5, 2027 compulsory-certification claim and its relevance to placements.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Marine Mechanical Technician — investigate a simulated service complaint

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Marine Mechanical Technician”, and have not been independently confirmed.

Supplied context: July 19–September 10, 2027; Trades; BCIT. Confirm the cohort's equipment and outcomes separately from the full foundation and apprenticeship pathways. **Integrated challenge:** Use a fictional service complaint, an approved manual extract and supplied test results to build an evidence-based troubleshooting record and customer handover. **Inputs to request:** Equipment-specific teaching extract, fictional work order, instructor-generated readings, parts information and an approved diagnostic sequence; no independent work on live systems.

Skill	Proposed activity → observable evidence
Reading	Locate the correct model's instructions and relevant diagram → references marked with model and document version.
Writing	Separate reported symptoms, supplied observations and suggested next checks → structured service note with no unsupported diagnosis.
Numeracy	Compare simulated readings with instructor-provided reference values → table with units, differences and clear flags for further review.
Digital	Navigate the assigned manual and enter a fictional service record → saved record that cites the exact source section.
Problem solving	Rank plausible explanations against the supplied evidence → justified next-check plan within the instructor's permitted sequence.
Communication	Explain findings to a fictional customer using plain language → explanation distinguishes what is known from what remains to be checked.
Collaboration	Divide document lookup, evidence recording and checking → shared record showing how conflicting interpretations were resolved.
Adaptability	Revise an explanation when new simulated evidence arrives → updated reasoning that explicitly addresses the new information.
Creativity and innovation	Design a clearer symptom-to-evidence recording aid → template that another learner can use accurately.

Accessible entry / stretch: Supply a labelled diagram, short manual extract and one symptom; extend to multiple possible causes or comparing two approved document revisions. **Evidence bundle:** Source annotations, readings table, troubleshooting record and customer-handover observation. **Research/provider questions:** Which engines and electrical/mechanical systems are taught? What equipment manuals and safe diagnostic procedures are authorized? How are workshop access, environmental handling and progression assessed?

Start from the Numeracy Hub

`numeracy-hub/index.html#mech` (gears, torque and hydraulics) and `#measure` (calipers, micrometers, gauges). Build the wider nine-skill task around a fictional service record.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Confirm whether the eight-week block is an introduction or part of a foundation pathway. Retrieve the current Marine Mechanical Technician outline/profile and equipment-specific teaching manuals; verify actual credit and prerequisites.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Security — observe, record and hand over a fictional incident

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Security”, and have not been independently confirmed.

Supplied context: September 14–November 6, 2027; Programs; provider TBD. Confirm the approved training provider and additional cohort content separately from licensing requirements. **Integrated challenge:** Observe a low-intensity scripted event, create a factual incident record and provide a respectful shift handover using a provider-approved response protocol. **Inputs to request:** Fictional site map, observation script, report template, communication protocol and authorized curriculum excerpts; use invented names and no real surveillance footage.

Skill	Proposed activity → observable evidence
Reading	Locate relevant instructions in fictional post orders and a site map → cited instruction and accurately identified location.
Writing	Produce an objective incident report → chronological account that separates direct observation, quoted statements and uncertainty.
Numeracy	Calculate elapsed time and reconcile event timestamps → timeline with checked intervals and consistent time notation.
Digital	Enter the report in a practice form and handle a fictional attachment → complete record with correct file naming and instructor-taught privacy handling.
Problem solving	Select the next step from provider-approved scenario options → reasoning tied to known facts, role limits and escalation instructions.
Communication	Practise respectful clarification and a concise radio-style handover → clear message without speculation or stigmatizing language.
Collaboration	Compare witness notes and reconcile differences without changing original observations → joint timeline retaining the source of each account.
Adaptability	Update the report when a new verified detail is supplied → dated addition that preserves the original account.
Creativity and innovation	Redesign a practice handover aid for readability → tested prompt sheet that improves completeness without changing approved procedures.

Accessible entry / stretch: Begin with still images, a short script and report headings; extend to several accounts, ambiguous details and a longer timeline. Allow opt-out from distressing role-play content. **Evidence bundle:** Incident report, timeline, digital practice record and handover observation. **Research/provider questions:** Which approved BST provider and assessment arrangements apply? What is covered in the remainder of the cohort? Which licensing steps, costs, technology and learner supports need separate planning?

Start from the Numeracy Hub

`source/book/chapters/ch01-whole-numbers.js` contains time-on-the-job and tables/graphs. Adapt the elapsed-time sequence to a factual incident log. Security law and procedures need a separate approved source.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Confirm the cohort provider and whether it uses the approved BST pathway. Independently check current JIBC/BC licensing guidance, assessment timing and completion evidence. Define the remainder of the longer cohort separately from BST theory.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Metal Fabrication — develop and check a pattern and parts plan

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Metal Fabrication”, and have not been independently confirmed.

Supplied context: September 14–November 6, 2027; Trades; BCIT. Confirm the actual fabrication scope, practical assessment and progression outcomes. **Integrated challenge:** Turn an instructor-approved simple drawing into a paper/card pattern, prepare a parts and materials list, and revise the layout after a specification change. **Inputs to request:** Approved drawing, material dimensions, pattern template, measurement tools and instructor-defined allowances; machine settings and physical fabrication remain partner-led.

Skill	Proposed activity → observable evidence
Reading	Interpret the drawing's dimensions, notes and part identifiers → annotated drawing linked to each pattern feature.
Writing	Record a clear sequence and inspection observations → work sheet that identifies what was checked and any unresolved discrepancy.
Numeracy	Calculate pattern dimensions and area using instructor-taught methods → labelled workings and measured pattern check.
Digital	Prepare or annotate a simple digital parts layout using available software → saved file with dimensions, units and revision label.
Problem solving	Find why the first pattern does not fit the model requirements → evidence-based correction using the drawing and approved allowances.
Communication	Explain a drawing ambiguity and request a precise clarification → question includes the part, dimension and conflicting information.
Collaboration	Exchange pattern and parts list for independent checking → recorded peer feedback and an agreed correction.
Adaptability	Revise the pattern when one specification changes → updated layout with affected dimensions identified.
Creativity and innovation	Compare two permitted nesting arrangements on a paper sheet → reasoned choice based on material use and clarity.

Accessible entry / stretch: Start with a simple flat shape, large dimensions and a partly labelled template; extend to a folded model or a more complex layout after instruction. **Evidence bundle:** Annotated drawing, pattern, parts list, calculation sheet and revision comparison. **Research/provider questions:** Which materials, machines and fabrication processes are included? Which drawing conventions and tolerances will be taught? Which references, consumables and assessment projects are supplied?

Start from the Numeracy Hub

`numeracy-hub/index.html#fab` (nesting, weight, square, bending and scale), `#coords` and `#measure` are the closest existing matches. Pair each calculation with a drawing, revision and peer-check record.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Confirm Metal Fabricator (Fitter) scope and level. Retrieve the current outline/profile and approved drawing conventions. Verify the actual student materials list rather than treating 2019 outline references as purchases.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Educational Assistant — support an accessible classroom activity

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Educational Assistant”, and have not been independently confirmed.

Supplied context: November 8–December 17, 2027; Programs; provider TBD. The supplied Douglas program is occupational context, not confirmation of the six-week cohort's credential or practicum. **Integrated challenge:** Using a fictional learner profile and teacher-provided goal, prepare an accessible classroom activity, collect neutral observations and suggest an adjustment for teacher review. **Inputs to request:** Fictional profile, teacher-directed goal, activity materials, observation form and approved privacy/role guidance; avoid real learner records or invented cultural information.

Skill	Proposed activity → observable evidence
Reading	Locate the teacher's goal, learner preferences and agreed supports → plan that accurately cites those details.
Writing	Record observable actions and the support offered → neutral notes that avoid diagnosis, judgement or unsupported interpretations.
Numeracy	Plan time and materials, then summarize fictional observation counts → checked schedule or table using appropriate units and clear labels.
Digital	Create an accessible visual schedule or practice resource → readable file with text alternatives and sensible document structure where supported.
Problem solving	Identify a barrier between the activity and stated goal → proposed adjustment that stays within the teacher-directed plan.
Communication	Model clear instructions and check understanding in a role-play → learner choice and comprehension are explicitly checked.
Collaboration	Coordinate with a role-play teacher and another support worker → agreed responsibilities and a concise handover.
Adaptability	Adjust pacing or presentation after the fictional learner's feedback → revised activity explaining the observed reason for the change.
Creativity and innovation	Offer two ways to engage with the same learning goal → practical alternatives that preserve access and learner dignity.

Accessible entry / stretch: Provide a completed example and short profile; extend to balancing two access needs or explaining why an appealing adaptation does not meet the stated goal. **Evidence bundle:** Activity plan, accessible resource, neutral observation note and proposed revision for teacher review. **Research/provider questions:** Who

delivers and recognizes this cohort? What practicum, screening and role expectations apply? Which inclusive-practice resources, assistive technologies and learner-support arrangements will be used?

Start from the Numeracy Hub

`source/book/chapters/ch01-whole-numbers.js` (time), `ch04-fractions.js` and `app-a-kitchen-math.js` offer general arithmetic structures. Replace trade context with teacher-directed classroom planning and source inclusion practices separately.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Identify the actual provider and recognized outcome. Check the selected partner's curriculum, placement arrangements and school-district expectations. Douglas is a reference example until chosen; confirm current open readings and accessibility resources.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

Nursing — organize a simulated care-information handover

All programs · Teaching framework · Worked numeracy starters

Research stage: Initial learning-design draft. Activities are proposed for ACCESS/instructor review. Cohort and public-program details come from the supplied reference, section “Nursing”, and have not been independently confirmed.

Supplied context: Dates TBD; Programs; BCIT. The supplied reference pathway is the BSN; confirm the ACCESS cohort's dates, preparation stage and teaching scope with BCIT. **Integrated challenge:** Use a fictional, instructor-authored patient case to organize information, check basic numerical records and provide a structured handover; identify gaps for a supervising instructor to resolve. **Inputs to request:** Fictional case, instructor-approved documentation and handover format, simulated chart and known-answer numeracy examples appropriate to the learners' training stage.

Skill	Proposed activity → observable evidence
Reading	Locate the case's relevant information, instructions and unknowns → annotated source and a list distinguishing known facts from missing data.
Writing	Draft a factual simulated note using the taught format → clear entry that avoids unsupported conclusions and preserves the source of information.
Numeracy	Check times, units and totals in an instructor-supplied simulated record → workings, units and independent check; flag discrepancies rather than infer treatment.
Digital	Enter information in a training chart or mock form → accurate record using fictional identifiers and approved privacy practices.
Problem solving	Notice conflicting or missing information and prioritize clarification → explicit account of the concern and appropriate instructor/supervisor escalation.
Communication	Give a structured handover and check understanding respectfully → concise exchange that includes the uncertainty and requested next step.
Collaboration	Practise complementary recorder, reviewer and communicator roles → jointly checked handover with individual contributions observed.
Adaptability	Revise the information summary after a new simulated observation → updated account that retains the original facts and records the change.
Creativity and innovation	Improve a plain-language teaching aid for the fictional case → prototype reviewed for accessibility and accuracy by the instructor.

Accessible entry / stretch: Start with short records, a vocabulary aid and one numerical check; extend to multiple documents and competing information needs. Medication calculations require a separate instructor-approved teaching and assessment sequence. **Evidence bundle:** Annotated case, simulated note, numerical checks, discrepancy log and observed handover; these do not establish clinical competence or licensure. **Research/provider questions:** What stage of BSN preparation is the ACCESS cohort addressing? Which current prerequisites and clinical-readiness deadlines apply? Which term-specific resources, simulation tools, cultural-safety learning and accessibility supports are approved?

Start from the Numeracy Hub

`source/book/chapters/ch05-decimals.js`, `ch06-percent-ratio.js`, `ch07-units.js` and chapter 1 time offer preparatory numeracy. Nursing faculty must approve clinical notation, methods and assessments.

These paths refer to the reference repository. See the source review for exact files, module identifiers and limitations. No new module is claimed to exist in that repository.

Next research action

Confirm the ACCESS delivery model and dates before scheduling. Verify BCIT prerequisites, recency rules, term-specific books and clinical-readiness deadlines. Use current BCCNM requirements for professional pathway information; do not infer a short nursing credential.

Record the exact source/version, relevant section, supported claim and ACCESS-specific applicability in the verification register. Select two or three focus skills per short lesson and collect evidence for all nine across the sequence.

IS4S: 11 worked numeracy starters

Back to the research pack

All quantities, schedules and records below are invented for learning. These are proposed preparation activities, not trade specifications, clinical guidance or credential assessments. Calculations use only the givens; real technical parameters and procedures come from the program's instructor. Each activity can be discussed aloud, drawn or completed with accessible calculation tools.

1. Welding — plan repeated blank lengths

- **Givens:** A paper planning task needs four blanks. Each requires 125 mm of usable length plus a fictional 3 mm planning allowance per blank. Available stock length: 600 mm. This allowance is a supplied exercise value, not a cutting standard.
- **Learner prompt:** How much total length should the plan allocate, and how much remains?
- **Worked answer:** One blank: 125 mm + 3 mm = 128 mm. Four blanks: 4 × 128 mm = **512 mm**. Remaining: 600 mm − 512 mm = **88 mm**.
- **Estimate/check:** Four usable lengths need 500 mm; four allowances add 12 mm, giving 512 mm. Check: 512 mm + 88 mm = 600 mm.
- **Common misconception:** Adding the 3 mm allowance only once gives 503 mm; “per blank” requires four allowances.
- **Scaffold / stretch:** Draw four labelled length bars. Stretch: six blanks require 768 mm, so the stated 600 mm is 168 mm short.
- **Next skill — Reading:** Underline the words that tell you how often to apply the allowance, then explain your interpretation to a partner.

2. Asphalt Paving — convert depth before finding volume

- **Givens:** A fictional rectangular classroom model represents an area 8 m long and 3 m wide, with a uniform depth of 50 mm. Ignore compaction, waste and all material properties for this exercise; 50 mm is not a recommended paving depth.
- **Learner prompt:** What geometric volume does the model represent, in cubic metres?
- **Worked answer:** Depth: 50 mm ÷ 1,000 = 0.05 m. Area: 8 m × 3 m = 24 m². Volume: 24 m² × 0.05 m = **1.2 m³**.
- **Estimate/check:** A depth of 0.05 m is one twentieth of a metre; 24 ÷ 20 = 1.2. Reverse check: 1.2 m³ ÷ 24 m² = 0.05 m.
- **Common misconception:** Multiplying by 50 without converting mm to m mixes units and makes the answer 1,000 times too large.
- **Scaffold / stretch:** Use a conversion strip showing 1,000 mm = 1 m. Stretch: at the same width and fictional depth, a 10 m length represents 1.5 m³.
- **Next skill — Digital:** Build a three-input spreadsheet with visible units and explain how you would detect an incorrectly entered depth.

3. Ironworker — reconcile an inventory record

- **Givens:** A fictional stores record lists 12 identical components, each with a supplied mass of 7.5 kg. These values are for inventory arithmetic only and provide no lifting or handling limit.
- **Learner prompt:** What total component mass belongs in the inventory record?
- **Worked answer:** 12 × 7.5 kg = (10 × 7.5 kg) + (2 × 7.5 kg) = 75 kg + 15 kg = **90 kg**.
- **Estimate/check:** Using 8 kg each gives 96 kg, so 90 kg is reasonable. Reverse check: 90 kg ÷ 12 = 7.5 kg each.
- **Common misconception:** Treating 7.5 kg as 7 kg and 5 g; the 0.5 kg is 500 g.
- **Scaffold / stretch:** Group the components into pairs, each totalling 15 kg. Stretch: if the count is corrected to 10, the total becomes 75 kg, a decrease of 15 kg.
- **Next skill — Collaboration:** One partner checks the count and another checks the supplied unit mass; agree on the corrected inventory entry and record who checked each input.

4. Carpentry — find a rectangle's diagonal

- **Givens:** A drawing shows a true rectangle with adjacent sides of 3 m and 4 m. The task is geometry practice; it does not establish construction tolerances or approval criteria.
- **Learner prompt:** What is the distance between opposite corners?
- **Worked answer:** The diagonal is the hypotenuse of a right triangle. $d^2 = (3\ m)^2 + (4\ m)^2 = 9\ m^2 + 16\ m^2 = 25\ m^2$. Therefore $d = \sqrt{25\ m^2} =$ **5 m**.
- **Estimate/check:** The diagonal must be longer than the 4 m side and shorter than the 7 m route along two sides. Also, $5^2 = 3^2 + 4^2$.
- **Common misconception:** Adding 3 m + 4 m finds the two-side route, not the direct diagonal.
- **Scaffold / stretch:** Draw and label the right triangle inside the rectangle. Stretch: doubling both sides to 6 m and 8 m doubles the diagonal to 10 m.
- **Next skill — Communication:** Explain why a 7 m answer represents a different path, using the drawing rather than only naming a formula.

5. Piping — combine straight segment lengths

- **Givens:** A classroom diagram contains three separate straight segments: 0.80 m, 1,250 mm and 0.65 m. Add only these stated lengths; the exercise includes no fittings, offsets, joining allowances or code requirements.
- **Learner prompt:** What is their combined length in metres and millimetres?
- **Worked answer:** 1,250 mm = 1.25 m. Total: 0.80 m + 1.25 m + 0.65 m = **2.70 m**, or **2,700 mm**.
- **Estimate/check:** Round to 0.8 m + 1.3 m + 0.7 m = 2.8 m, close to 2.7 m. Separate check in millimetres: 800 + 1,250 + 650 = 2,700 mm.
- **Common misconception:** Moving the decimal only two places converts 1,250 mm to 12.5 m; millimetres to metres requires division by 1,000.
- **Scaffold / stretch:** Rewrite all three values in one unit before adding. Stretch: two identical sets total 5.40 m.
- **Next skill — Writing:** Produce a short material-length note with a clear heading, one consistent unit and the calculation used.

6. Tower Crane — standardize a fictional load record

- **Givens:** A classroom record has three mass entries: 750 kg, 0.45 metric tonnes and 125 kg. One metric tonne (t) equals 1,000 kg. This is record-conversion practice only; the record is incomplete for real lifting and establishes no crane capacity, safe load, rigging choice or lift authorization.
- **Learner prompt:** Rewrite the entries in kilograms and total them, then express the total in tonnes.
- **Worked answer:** 0.45 t × 1,000 kg/t = 450 kg. Total: 750 kg + 450 kg + 125 kg = **1,325 kg**. In tonnes: 1,325 kg ÷ 1,000 kg/t = **1.325 t**.
- **Estimate/check:** The first two entries total 1,200 kg; adding 125 kg gives a little over 1.3 t. Reverse check: 1.325 t × 1,000 kg/t = 1,325 kg.
- **Common misconception:** Reading 0.45 t as 45 kg produces a tenfold conversion error for that entry.
- **Scaffold / stretch:** Make a two-column “given unit / kilograms” table. Stretch: if the 125 kg entry was a duplicate record, the corrected total is 1,200 kg or 1.2 t.
- **Next skill — Problem solving:** Explain why a correct sum alone cannot answer whether any crane can lift these items; list the missing information you would refer to the qualified instructor.

7. Marine Mechanical Technician — calculate elapsed workshop time

- **Givens:** An invented workshop observation log starts at 09:40 and ends at 10:25 on the same day, with no breaks. No engine or equipment performance specification is implied.
- **Learner prompt:** How long did the observation last, in minutes and decimal hours?
- **Worked answer:** From 09:40 to 10:00 is 20 minutes; from 10:00 to 10:25 is 25 minutes. Total: 20 + 25 = **45 minutes**. Decimal hours: 45 ÷ 60 = **0.75 hours**.
- **Estimate/check:** The interval is less than an hour and equals three quarters of an hour. Check: 09:40 + 45 minutes = 10:25.
- **Common misconception:** Subtracting 9.40 from 10.25 as decimal numbers gives 0.85; clock minutes use groups of 60, not 100.
- **Scaffold / stretch:** Mark the interval on a clock or number line with 10:00 as a bridge. Stretch: an end time of 11:15 would give 95 minutes, or 1 hour 35 minutes.
- **Next skill — Adaptability:** Update the log when the supplied end time changes, and explain which part of your original calculation remains useful.

8. Security — read a shift schedule

- **Givens:** An invented same-day schedule runs from 14:30 to 22:00 and includes a 30-minute break from 18:00 to 18:30. This exercise asks for scheduled time excluding the break; it makes no claim about paid time, employment rules or required breaks.
- **Learner prompt:** How many hours and minutes remain after subtracting the listed break?
- **Worked answer:** 14:30 to 22:00 is 7 hours 30 minutes, or 450 minutes. Subtract 30 minutes: 450 − 30 = **420 minutes**, or **7 hours**.
- **Estimate/check:** 7.5 hours − 0.5 hours = 7 hours. Add back the break: 420 + 30 = 450 minutes.
- **Common misconception:** Writing 7 hours 30 minutes as 7.30 decimal hours; 30 minutes is 0.5 hours.

- **Scaffold / stretch:** Split the timeline at 18:00 and 18:30. Stretch: ending at 22:15 with the same break produces 7 hours 15 minutes excluding the break.
- **Next skill — Writing:** Write a factual two-sentence schedule note that distinguishes start/end times, the break and the calculated time, without adding any invented incident or personal information.

9. Metal Fabrication — calculate material utilization on paper

- **Givens:** A paper model of a rectangular sheet measures 120 cm × 80 cm. Six rectangles measuring 40 cm × 20 cm are placed on it in a three-column, two-row arrangement. Assume zero cutting loss for this paper exercise; these are not workshop cutting instructions.
- **Learner prompt:** What area do the six rectangles occupy, and what percentage of the sheet is that?
- **Worked answer:** Sheet area: 120 cm × 80 cm = 9,600 cm². One rectangle: 40 cm × 20 cm = 800 cm². Six rectangles: 6 × 800 cm² = **4,800 cm²**. Utilization: 4,800 ÷ 9,600 × 100% = **50%**. Unused area: **4,800 cm²**.
- **Estimate/check:** The layout occupies the full 120 cm width and 40 cm of the 80 cm height, exactly half the sheet. Used plus unused area: 4,800 + 4,800 = 9,600 cm².
- **Common misconception:** Adding length and width finds neither area nor utilization; area requires multiplication and square units.
- **Scaffold / stretch:** Draw the sheet as a 3 × 4 grid of equal rectangles and shade six cells. Stretch: nine cells use 7,200 cm², or 75%.
- **Next skill — Creativity and innovation:** Draw another arrangement of the six rectangles, then explain how you know it fits and has the same occupied area.

10. Educational Assistant — interpret a visual activity schedule

- **Givens:** An invented group activity schedule starts at 09:00: welcome 10 minutes, reading 15 minutes, making 20 minutes, break 15 minutes, in that order. It contains no real learner information and makes no recommendation about an individual's support needs.
- **Learner prompt:** What fraction of the one-hour schedule is reading, and when does making begin?
- **Worked answer:** Total: 10 + 15 + 20 + 15 = **60 minutes**. Reading fraction: 15/60 = **1/4**, or **25%**. Welcome ends 09:10; reading ends 09:25, so making begins at **09:25**. The full schedule ends at **10:00**.
- **Estimate/check:** Four 15-minute blocks fill an hour. Check the timeline: 09:25 + 20 minutes = 09:45, then + 15 minutes = 10:00.
- **Common misconception:** Confusing an activity's duration with its clock start time, or using 15/100 because an hour is mistaken for 100 minutes.
- **Scaffold / stretch:** Show four quarter-hour blocks beside the activity timeline. Stretch: transfer five minutes from making to reading; reading becomes 20/60 = 1/3 of the unchanged hour.
- **Next skill — Communication:** Explain the schedule using both words and a visual sequence, then invite a partner to show when the next activity begins.

11. Nursing — total a synthetic intake practice record

- **Givens:** A fictional arithmetic record lists 250 mL, 0.15 L and 100 mL, all marked as measured intake volumes in the exercise. There is no real patient, fluid target, medication or clinical decision. One litre equals 1,000 mL.

- **Learner prompt:** Rewrite all entries in millilitres and find the total in millilitres and litres.
- **Worked answer:** 0.15 L × 1,000 mL/L = 150 mL. Total: 250 mL + 150 mL + 100 mL = **500 mL**. In litres: 500 mL ÷ 1,000 mL/L = **0.5 L**.
- **Estimate/check:** 250 mL is one quarter of a litre; 150 mL + 100 mL adds another quarter, making half a litre. Independent unit check: 0.25 L + 0.15 L + 0.10 L = 0.50 L.
- **Common misconception:** Treating 0.15 L as 15 mL; it equals 150 mL. Do not infer clinical appropriateness from the total.
- **Scaffold / stretch:** Use a 1 L diagram divided into ten 100 mL sections. Stretch: an additional fictional 0.20 L entry brings the total to 700 mL or 0.7 L.
- **Next skill — Digital:** Enter the synthetic values into a practice table with separate value and unit fields; compare the result with the hand calculation before marking it checked.

Arithmetic verification

All 11 examples, including their numerical stretch answers, were checked with exact Python fraction/integer arithmetic in the arithmetic checker. The checker verifies calculations, not real-world suitability, occupational standards or clinical practice. Instructor review is still needed to select the appropriate learning level and context.

Make, measure, improve

Paper-glider lab: an introduction to the nine skills

Back to the research pack

Proposed 60-minute pilot · Adult foundational learning · Instructor review before delivery

Purpose: Follow a visual instruction, collect useful measurements, compare results and explain an improvement. Connect the work to reading a job sheet, recording a measurement and sharing a change with a team.

Source connection: The three supplied ACCESS Essential Skills lesson images use a delta-wing paper-plane task, instructor/learner activity columns, pre-assessment, hands-on work and reflection. This is a new adaptation. The fourth page and the original folding handout were not supplied. The glider below is a simple classroom model, not a reconstruction of that missing handout or an aerodynamics lesson.

Focus for feedback: Reading, Numeracy and Communication. Collect brief evidence for the other six skills across the sequence. The learner's method and improvement matter more than distance travelled.

What you need

- Two sheets of ordinary paper per pair, a ruler or tape, pencils and the record sheet below.
- A clear, instructor-controlled testing lane with a marked start line and soft target zone. Use paper only; keep people out of the lane. Launch only on the facilitator's signal and retrieve together.
- One accessible device per pair with a spreadsheet or simple table tool. A calculator is welcome.
- A demonstrated model and enlarged numbered instructions. Offer a partner-assisted or facilitator-launched version so learners can direct, observe and measure without throwing.

If a device is unavailable, complete the record on paper and schedule the digital entry later. Mark Digital **not observed yet**, rather than treating a paper table as digital evidence. For a larger class, use several supervised lanes or extend the testing time.

Make the simple model

The facilitator demonstrates each step, then checks that learners can identify the next step in the numbered instructions.

1. Place a rectangular sheet with a short edge at the top. Fold the left long edge to meet the right long edge. Crease, then unfold to show the centre line.
2. Fold the two top corners toward the centre line so their top edges meet at the line, making a pointed top.
3. Fold each new sloping outer edge toward the centre line to make a narrower point.
4. Fold the model in half along the centre line, keeping the folded layers on the outside.
5. Make one wing by folding a top sloping edge down toward the long bottom edge. Turn over and repeat on the other side.
6. Open the wings gently so the body can be held underneath. Compare the two wings. Point out any difference before testing.

Use the first model to practise following and clarifying instructions. If a fold is difficult, demonstrate again or use a prepared model while the learner directs the steps. Test the activity locally before teaching; the outcomes are following, measuring and improving, not a promised flight performance.

Session plan

Time	Instructor actions	Learner actions	Resources / evidence
0–5 min	State the first goal: explore how far the model travels within the testing lane. Introduce the task. Ask where learners already measure, follow instructions or explain a process. Invite examples without requiring personal disclosure.	Name a familiar strategy and predict how far the model might travel.	A baseline prediction with a unit.
5–10 min	Demonstrate the numbered folds and establish the launch/measurement routine. Model how to ask for clarification.	Read or track the steps, identify a confusing instruction and check understanding with a partner.	Numbered instructions; annotated or explained step.
10–18 min	Support making and purposeful role sharing.	Make the model, write one prediction and agree launcher/director, measurer and recorder roles. A pair can combine roles and swap.	Model, prediction, role note.
18–28 min	Supervise three baseline trials per model. Demonstrate measuring along the lane from the start line to the first landing point's position along the lane.	Estimate, measure and record three distances to the nearest 10 cm. Keep start position and launch routine consistent.	Three labelled baseline measurements.
28–36 min	Model a table and a mean with the fictional data below. Explain the difference between an estimate and a measured result.	Enter their own results digitally, calculate a mean and independently check the total.	Saved table, units and calculation.
36–44 min	Introduce the changed goal: aim for a clearly marked target zone rather than the longest flight.	Propose one model change, explain the reason and agree how to check it. Record the change before testing.	Change sketch and revised plan.
44–52 min	Supervise three revised trials under the same measurement routine.	Test, record and compare. Discuss variation and whether the revision helped the new goal.	Revised data and a cautious evidence-based conclusion.
52–60 min	Invite a short partner handover and exit response. Connect strategies to a selected program brief.	Explain a finding, ask/check one clarification, identify skill evidence and choose a next practice step.	Handover observation and exit note.

Learner record

Names or learning IDs: _____ Date: _____ Roles: _____

Our first prediction: _____ cm. **Reason: _____**

Trial	Baseline distance (cm)	Revised distance (cm)	Observation
1			
2			
3			
Total			
Mean: total ÷ 3			

The target zone: _____ to **_____** cm along the lane.

One change we made: _____

Why we chose it: _____

What we kept the same: _____

How many trials landed in the target zone before / after: ____ / ____

What the results suggest, and what we would check next: _____

Worked example — invented data

The practice target is **170–210 cm inclusive** along the lane. Baseline trials are **120, 150 and 180 cm**. Their total is `120 + 150 + 180 = 450 cm`. The mean is `450 ÷ 3 = 150 cm`, or **1.50 m**. One of three trials is in the target zone.

After one change, fictional trials are **150, 180 and 210 cm**. Their total is `540 cm`; the mean is `540 ÷ 3 = 180 cm`, or **1.80 m**. Two of three trials are in the target zone. The mean increased by **30 cm**.

Calculator steps: `120 + 150 + 180 =`, then `÷ 3 =`. Repeat with the revised data. Clear between separate calculations as needed. These are ordinary operations, not verified model-specific key recordings.

Spreadsheet steps: Put the three baseline values in cells B2:B4 and revised values in C2:C4. In B5 enter `=SUM(B2:B4)` and in B6 enter `=AVERAGE(B2:B4)`; repeat for column C. Label the columns **cm** and check the spreadsheet against the hand calculation. Formula names may differ with the tool's language settings.

Check: Each mean must lie between the smallest and largest trial. Three attempts provide limited evidence: changes in launching and measurement can affect the results. A longer mean is not automatically better for a target-zone goal.

Make the nine skills visible

Skill	Task evidence	Instructor look-for
Reading	Numbered instruction identified and applied.	Learner finds the next step and uses the instruction to check the model.
Writing	Prediction, change note and short conclusion.	Another person can understand what was tried and observed.
Numeracy	Labelled measurements, total, mean and target count.	Consistent units, sensible method and a reasonableness check.
Digital	Table created and saved with checked values/formulas.	Learner can find the saved result and explain one entry or formula.
Problem solving	Explanation of inconsistent results or a misleading mean.	Learner considers evidence and identifies a useful next check.
Communication	Partner handover plus clarification/check-back.	Listener accurately restates the finding or asks a relevant question.
Collaboration	Agreed roles and shared checking.	Both learners contribute and use feedback constructively.
Adaptability	Plan revised after the goal changes.	Learner explains why the original strategy may no longer suit the target.
Creativity and innovation	One proposed modification, tried and evaluated.	Learner connects an idea to a purpose and reflects on the test.

Exit prompts and transfer

1. Point to one result you checked. How did you know it was reasonable?
2. Show where you used a skill other than Numeracy. What would an observer see?
3. What would you change or keep for the next attempt, and why?

Choose a transfer task: a **welding cut list**, **asphalt quantity table**, **security timeline**, **classroom activity schedule**, or **synthetic nursing volume record**. Ask which checking strategy carries over and which new source or instruction is needed. Use the corresponding program brief and numeracy starter.

Before the pilot

Prepare a model and verify the folds, room layout and measurement routine; set a target suited to that space. Confirm device availability and supports. After the pilot, record actual timings, unclear instructions and learner feedback before expanding this into program-specific lessons. This lesson has been designed and its example arithmetic checked; it has not been classroom-tested.

IS4S research source audit — working notes

Prepared 22 October 2026 from the user-supplied 2027 planning text and three photographed lesson pages. This audit checks internal consistency and instructional design; it does not independently verify external program or regulatory claims.

What the supplied sources establish

- The planning document lists 11 tentative cohorts: Welding, Asphalt Paving, Ironworker, Carpentry, Piping, Tower Crane, Marine Mechanical Technician, Security, Metal Fabrication, Educational Assistant and Nursing.
- Dates, partners and dedicated ACCESS cohort arrangements are supplied planning context, not independently confirmed bookings.
- The document itself distinguishes short ACCESS cohort dates from longer public foundation, apprenticeship and degree pathways. Preserve that distinction in every program page.
- The pasted source retains descriptive source labels but no URLs. Its statement that research was checked on 21 October, with Nursing added 22 October 2026, describes the supplied document's provenance; it is not evidence of a new verification by this research project.
- The photographs show pages labelled 1/4, 2/4 and 3/4 of an ACCESS Essential Skills lesson. Page 4 and the separate Delta Fighter folding instructions are absent. Do not describe the lesson as fully reviewed or reconstruct missing instructions as original content.
- The pasted planning document ends mid-word at “deliv”. The final staff checklist is incomplete.

Planning issues to resolve

Issue	Evidence in supplied text	Required resolution
Tower Crane start-date conflict	Main table and program section say 5 July 2027; summer paragraph says Tower Crane and Marine both begin 19 July.	Confirm the intended Tower Crane date and update the summer paragraph from the controlled schedule.
Partner spelling and identity	Tower Crane table says “IOUE Maple Ridge”; narrative suggests IUOE Local 115.	Confirm the partner's legal/display name and site. Do not silently convert a suggestion into confirmation.
Status field is missing	Instructions say to update a Status field; the visible table has no Status column.	Add a status column in the maintained schedule and retain Tentative until confirmed.
Unidentified providers	Asphalt Paving has a duration in its partner field; Security and Educational Assistant say TBD.	Confirm providers before treating IUOE, JIBC or Douglas reference programs as the actual cohort curricula.
Unusual autumn date pattern	Security and Metal Fabrication run Tuesday 14 September to Saturday 6 November 2027.	Confirm that the Tuesday start and Saturday finish are intentional; calendar span alone does not establish attendance days.
Nursing delivery model and dates	Dates are TBD; the reference is a three-year BSN while most other rows describe six- or eight-week cohorts.	Confirm whether ACCESS is supporting degree entry, prerequisite preparation, a bridge, or another defined block, plus its delivery dates.

Issue	Evidence in supplied text	Required resolution
Course scope and credit are unspecified	Several eight-week cohorts are compared with 21–28-week foundations; Tower Crane technical levels exceed the listed cohort span.	Obtain each cohort's course code, training level, contact hours, assessed outcomes and credit agreement. Do not imply full foundation or apprenticeship completion.
Resource conflicts need an owner	Several cohorts run concurrently and use common departments or partners.	Check rooms, instructors, learner supports, equipment and transport against the actual timetable. Overlap is a coordination flag, not proof of a conflict.

Overlaps visible in the tentative dates: Welding/Asphalt, Welding/Ironworker during 15–19 March, Ironworker/Carpentry, Piping/Tower Crane during 5–9 July, Tower Crane/Marine, and identical Security/Metal Fabrication dates. Nursing cannot yet be included in capacity planning.

Verification priorities

Priority 1 — before recruitment or any claim about eligibility/credit

1. Obtain partner confirmation of the ACCESS-specific offering: course identifier, location, dates, hours, admission/equivalency process, credential or credit, and included assessments. Dedicated seats do not imply waived academic or professional requirements.
2. Verify the stated 5 July 2027 Tower Crane compulsory-certification change against current SkilledTradesBC guidance and clarify its application to the intended learner placements. The supplied text gives this date, but this audit has not independently confirmed it.
3. Resolve Generalist versus Reinforcing Ironworker and the actual Plumbing/Steamfitting/Sprinklerfitting coverage within Piping. These are not interchangeable qualifications.
4. Verify Nursing academic preparation, anatomy/physiology recency, clinical-readiness and licensure requirements with BCIT/BCCNM. Avoid copying grade thresholds, fees or deadlines into learner recruitment content until checked.
5. Verify Security training, assessment and licence requirements separately. Provider training completion is not permission to work.

Priority 2 — before curriculum mapping and materials production

- Retrieve the current ESDC Skills for Success framework definitions and use its nine current skill names consistently.
- Retrieve the exact current SkilledTradesBC outline/profile for each selected trade and training level; record effective date/version and specific competency identifiers used in lesson mapping.
- Obtain provider-approved assessment criteria and authentic, reusable sample documents: drawings, work orders, material lists, schedules, logs, inspection forms, incident reports, learning-support plans and de-identified clinical simulation records.
- Completed source/content inspection of the Numeracy Hub at commit `7c0ceaab485e36aa45c596ac58e43bba63e4f824`; see the Hub review. Browser usability and accessibility testing remain separate work.

- Verify public reference-program claims most likely to change: IUOE level structure, JIBC assessment/completion rules, BCIT program prerequisites, and Douglas program/practicum requirements.
- Confirm Indigenous learner-support and cultural-content approaches with ACCESS and relevant community partners. Do not invent Nation-specific teachings or use cultural content as decorative theming.

Priority 3 — before purchasing or delivery

- Obtain dated instructor-approved books, editions, access codes and supplied-versus-purchased items for the actual cohort. The supplied references from 2019–2026 are background, not a 2027 shopping list.
- Confirm code/manual editions with the provider and applicable current standards. Distinguish an exam's permitted resource edition from rules applicable to work.
- Confirm PPE, equipment, accessibility, digital access, transport, placement support, and provider-controlled safety/clinical arrangements.

Historical Essential Skills and current Skills for Success

Use the current nine skill names for new IS4S resources. Retain historical labels when describing the archived lesson. This crosswalk is a design aid, not a one-to-one official equivalence table.

Historical Essential Skills term	Treatment in new resources
Reading / Reading Text	Reading: locating, interpreting and using written information; demonstrate understanding through the task.
Document Use	Embed across Reading, Numeracy, Writing and Digital according to the document task: interpreting a chart, extracting a measurement, completing a form or navigating a digital record. Keep document literacy visible.
Writing	Writing: producing useful, accurate messages, forms, logs and explanations for a specified audience.
Numeracy	Numeracy: selecting and using numbers, measurements, quantities, time and data, then checking whether the result makes sense.
Oral Communication	Communication: understanding, exchanging and checking information through appropriate spoken, nonverbal and other modes.
Thinking	Problem solving captures much reasoning and decision-making; some tasks also involve Creativity and innovation or Adaptability. Map the actual behaviour.
Working with Others	Collaboration: agreeing roles, coordinating work, exchanging feedback and contributing to a shared result.
Computer Use	Digital: finding, creating, managing and communicating information with digital tools; broader than basic computer operation.
Continuous Learning	A practice running across all nine: reflect, seek feedback, practise and transfer. Adaptability may be demonstrated when a learner changes approach, but it is not simply a renamed Continuous Learning category.

Historical Essential Skills term	Treatment in new resources
No separate historical category	Creativity and innovation is an explicit current skill: generate, test and improve a useful idea within the task's constraints.

Current nine: Reading; Writing; Numeracy; Digital; Problem solving; Communication; Collaboration; Adaptability; Creativity and innovation.

What to retain from the photographed lesson

- An inviting, concrete opening; clear purpose; a brief pre-assessment; hands-on practice; learner explanation; a short post-assessment; and a summary.
- Separate instructor and learner activity columns, with resources and estimated time.
- A task that gives learners something visible to make, try, discuss and improve.
- Questions that help learners name skills they already use and transfer them to another situation.
- A practice/debrief cycle rather than a long abstract introduction to skill definitions.

What to strengthen for a new nine-skills version

- The original visible activity primarily supports following a folding document, problem solving, discussion and turn-taking. It does not visibly generate performance evidence for all nine current skills. Merely naming a skill during the debrief is not evidence of using it.
- Make numeracy observable: estimate flight distance, measure repeated trials with a stated unit, record results in a table, compare results and explain variation. Use measurements appropriate to learners' readiness.
- Add a short written prediction and results note; a simple digital table or graph where access permits; a paired explanation with a comprehension check; shared testing roles; an announced condition change; and a learner-designed revision with a reason.
- Assess a few focus skills per lesson and rotate emphasis across the program. All nine can be visible across a learning sequence without making every short task carry nine equally weighted assessments.
- Use task evidence, a brief learner explanation and instructor observation together. A strong written answer should not mask missing practical understanding; reading or writing support should not be misread as lack of reasoning.
- Offer accessible materials, large-print/visual instructions, demonstration, assistive tools and varied response modes while preserving the skill being assessed. Record the support used where it affects interpretation.
- Use a controlled launch area, suitable materials and alternatives to throwing. Reconsider the original prize/competition format: collaborative accuracy or improvement can support participation without requiring a public ranking.
- Keep the historical aircraft narrative as source content pending fact-checking, or replace it with a short verified context. Do not turn an unverified aerospace claim into a new learning objective.

Source-register fields for the next research pass

For every substantive claim record: claim ID; program; exact claim; source title and URL; issuing organization; publication/effective date; access date; page/section; applicability to public reference program versus ACCESS cohort; verification status; and the learner-facing consequence. Suggested statuses: supplied only, primary source checked, partner confirmation pending, conflicting sources, historical reference, proposed learning activity.

Keep proposed activities, authoritative occupational requirements and partner-specific decisions visibly distinct. Learners should be able to see what they will practise; staff should be able to trace why it was selected.

Sources and the next verification pass

Back to the research pack

Research record · 22 October 2026

This pack begins with direct review of the supplied material and the Numeracy Hub. It does not certify the supplied document's claims as current. A source label without a URL, edition or page is a research lead, not a retrievable citation.

Material inspected

ID	Source	What was actually checked
S01	User-supplied planning reference	All 11 program sections, schedule, claimed prerequisites/pathways, books and staff notes. The source says checked Oct 21 and Nursing added Oct 22, 2026; those are its stated dates, not a new independent verification. The final sentence is truncated.
S02	Three photographed ACCESS Essential Skills lesson pages supplied in chat	Visible pages 1–3 of 4: opening, pre-assessment, folding task, discussion, teamwork, post-assessment and summary. Page 4 and separate folding instructions were unavailable.
H01	Numeracy Hub repository at the inspected commit	Repository retrieved and source/content inspected. See Hub review for file/line citations. No browser or classroom validation claimed.
H02	Hub brand guidance	Current nine skill names, design colours, recommended fonts, badge/lockup rules and numeracy teaching context.
D01	Program briefs, numeracy examples and pilot lesson in this pack	Newly proposed curriculum content. Fictional examples are clearly labelled; arithmetic verification does not validate occupational or clinical practice.

Official source destinations

The following links are **starting points for verification**, not citations establishing the program claims. Except for the Skills for Success URL already present in H01, these are organization-level entry points; the next pass must capture the exact program page or document and its effective date.

Source / destination	Program relevance	Access result this session
ESDC Skills for Success	Current skill definitions, components and proficiency guidance	Proxy 403
SkilledTradesBC	Welder, Asphalt Laydown Operator, Ironworker, Carpenter, piping trades, Tower Crane, Marine Mechanical Technician, Metal Fabricator	`/welder` request: proxy 403

Source / destination	Program relevance	Access result this session
BCIT	Actual reference programs, prerequisites, clinical readiness and provider contacts	Proxy 403
BCIT Library guides	Dated Nursing booklists and resources	Proxy 403
UAPICBC	Welding and Piping public-program context; actual cohort scope through partner confirmation	Proxy 403
IUOE Local 115	Asphalt and Tower Crane reference pathways; verify partner/site identity	Proxy 403
JIBC	Basic Security Training curriculum, assessment and completion evidence	Proxy 403
Douglas College	Educational Assistant reference program and current open readings	Proxy 403
BCCNM	Current RN competencies, standards and registration pathway	Proxy 403
WorkSafeBC	Current B.C. workplace requirements supporting provider-led trade instruction	Proxy 403

“Proxy 403” describes this environment's request failure. It is not evidence that the source is unavailable publicly, the organization denied the user access, or the underlying program claim is false.

Additional teaching-resource leads from the Hub

See the Hub's extracted source list: BCcampus open trade mathematics and pattern development, NIST conversion factors, the B.C. plumbing-code bulletin, calculator instructions and measuring-instrument guides. These links were found in repository content; their current pages and reuse terms have not been checked in this session. Validate those before quoting or adapting source material. The repository's NPC 2025 content does not itself establish the code applicable to a B.C. cohort.

Verification queue

Open the CSV register. Each row identifies a material claim or decision, its provenance, the exact question to resolve and its status. Append the retrieved document URL, effective date and page/section when checked. A provider decision cannot be resolved solely by a public website.

Use these statuses consistently:

- **Supplied only:** recorded in S01 but not independently checked.
- **Repository checked:** directly observed in the specified H01 commit.
- **Conflicting supplied information:** two parts of S01 disagree.
- **Partner confirmation pending:** an ACCESS-specific arrangement that the selected provider must confirm.
- **Proposed learning activity:** original teaching design awaiting review/pilot.
- **Primary source checked:** use only after retrieving the exact current source and recording what it supports. None of the external program requirements in this first pack have this status.

Network draft saved

A configuration draft was saved with the existing package-manager/GitHub network preset preserved and these custom domains added: `www.canada.ca`, `skilledtradesbc.ca`, `www.bcit.ca`, `libguides.bcit.ca`, `uapicbc.ca`, `www.uapicbc.ca`, `www.iuoe115.ca`, `iuoe115.ca`, `www.jibc.ca`, `www.douglascollege.ca`, `www.bccnm.ca`, `www.worksafebc.com`.

The tool confirmed the draft was saved and requires publication. Saving the draft did not activate the rules. Review and save the changes in environment settings, then publish the environment; after activation, retry the exact sources and complete the verification queue. Additional teaching-resource hosts above and B.C. government licensing pages may need narrow additions when their exact required links are established. No secrets or credentials are required for these public-source checks.

Prepared from the supplied planning reference, three photographed lesson pages and the Numeracy Hub source. External program and regulatory verification remains pending. This workbook is a generated review copy of the editable Markdown research pack.